S(12): A COMPUTER-ASSISTED DETERMINATION

Theorem:
 S(12) = {2,3,5,7,11,13,17,19,23,29,31,37,43}.

All computational obligations below have completed. The 656 required character orders, including the 53 reused audited certificates, have coprime Pade certificates verified in exact finite-field arithmetic. The final coverage check accounts for all 44045 primes up to 532900. The new local computations cover degree 12 and include levels 107 and 109. Earlier audited results are identified explicitly.

1. OCCURRENCE AND REDUCTION TO DEGREE-12 POINTS

The published sextic classification gives S(6)={2,3,5,7,11,13,17,19,37}. A quadratic extension of a sextic field gives a field of exact degree 12 and preserves its torsion points. For 23,29,31, the script verify_small_witnesses.m constructs degree-12 number fields, proves their defining polynomials irreducible, and checks nonzero points of the indicated prime order on nonsingular elliptic curves. The exact coefficients are recorded in small_witnesses.json.

For 31 the specialization is obtained from the degree-12 function
 (x^2*y-x*y^2+y-1)/((x-y)*x^2*y)
on X_1(31), given by Derickx--van Hoeij, Section 2.2. An explicit irreducible degree-12 fiber and its elliptic curve are checked; the occurrence argument does not rely solely on Hilbert irreducibility.

For 43, let a be a root of
 f=t^12-t^11-5t^10+6t^9+9t^8-16t^7-3t^6+19t^5
       -3t^4-12t^3+t^2+4t+1,
and put x=a and
 y=(35a^11-65a^10-107a^9+314a^8-34a^7-580a^6
       +601a^5+156a^4-552a^3+219a^2+136a-81)/43.
Set r=(x^2*y-x*y+y-1)/(x^2*y-x), s=(x*y-y+1)/(x*y), b=r*s*(r-1), c=s*(r-1). Then on
 Y^2+(1-c)XY-bY=X^3-bX^2
the point (0,0) has order 43. occurrence43.m verifies irreducibility, nonsingularity, and the point order exactly. This is a van Hoeij low-degree-place entry; its j-invariant has degree 4.

Conversely, a prime dividing E(K)_tors occurs as the order of a point by Cauchy’s theorem. For p>37 and [K:Q]=12 the corresponding closed point on X_1(p) has degree dividing 12. The known classifications in degrees 1,2,3,4,6 exclude every proper divisor. Thus its degree is 12. Write D for its irreducible rational trace divisor. This is precisely the S_new(12) condition in Derickx--Stoll, Definition 7.1.

2. GLOBAL PRIME BOUNDS

Oesterle’s effective bound, as proved in DKSS, gives
 p <= (3^(12/2)+1)^2 = 532900.
The published Derickx--Stoll strange-character Table 2 is used only for p<100000. It is not extrapolated.

For 149<=p<100000, Corollary 7.5 excludes all 9558 primes: its small-prime alternative fails since p^2>(2^19/325)*12+1, its CM alternatives would give p=25 or 37, and its remaining alternative requires (p-1)/(2*str(p)) to divide 12. No table entry or non-strange

prime in this interval satisfies that divisibility. The complete integer checks are in published_prime_bound.json.

For p=113,127,131,137,139 the prime is non-strange, and
gon_Q(X_1(p)) >= 325*(p^2-1)/65536 > 60.
Derickx--Stoll Proposition 4.1, with diamond <2>, ell=2, n=5, applies; Remark 4.4 supplies its rational 2-primary torsion hypothesis. It makes D a union of diamond <2> orbits. The orders in the diamond group are respectively 14,7,65,34,69. None of these orders, or their possible integral halves or thirds, divides 12. This is impossible for the irreducible divisor D.

For 100000<p<=532900 we use only odd-characteristic winding-quotient formal immersion. Let e be the X_0(p) winding element and put
T'_n=sum_{m|n} mu(n/m)T_m, L_n=T'_(2n)-2T'_n.
The integral spans of L_1,...,L_12 and (T_2-3)T_1,...,(T_2-3)T_12 coincide. DKSS's intersection formula, for odd M, p>24M and pu=1 modulo M, is
L_n e . {0,a/M}=epsilon_M(na)-epsilon_M(nu/a), 1<=n<=12.
Here epsilon_M is the indicator of the upper half of the nonzero residues. We verify rank 12 for every unit u, using
ell=3,M=61 (60 matrices), ell=5,M=59 (58 matrices), and
ell=7,M=59 (58 matrices).
As (T_2-3)Te lies in integral homology, these are integral independence certificates for Parent's odd-characteristic formal immersion criterion at 12 times the relevant rational cusp. Rational torsion on the winding quotient injects under reduction at these odd primes.

Consequently, at every ell in {3,5,7}, for at least one residue degree 1<=f<=12, either the Hasse interval for ell^f contains a multiple of p, or p divides ell^f-1 or ell^f+1. Otherwise prime torsion cannot be in good or toric reduction, additive reduction cannot support p>100000, and all relevant modular points specialize to the common cusp type on X_0(p), contradicting formal immersion. An Atkin--Lehner involution interchanges the two cusp conventions if necessary.

We sharpen the Hasse test when the trace is divisible by ell. Such an elliptic curve is supersingular. For ell=3,5 the only supersingular j is 0, and for ell=7 it is 1728. These statements follow directly from the Hasse invariant: the relevant coefficients are a2 for a characteristic-3 equation y^2=x^3+a2*x^2+a4*x+a6, 2A for y^2=x^3+A*x+B in characteristic 5, and 3B in characteristic 7. ss_orders.m enumerates all twists of these curves over each relevant finite field. No other group order is accepted for a supersingular trace. A condition retained by this filter need not be sufficient; only necessity is used.

Of the 34453 primes in this interval, these tests exclude 34361 and leave 92. For each survivor let m=(p-1)/2. We choose a prime r with a=v_r(m)>b=v_r(12), and test every even character whose order is a multiple of r^(a-b). There are 656 required tests: 53 are reused from the independently audited degree-11 computation and 603 are new. The selections and all divisors to be tested are explicit in large_selected_orders.json.

For a tested character chi, over an exact auxiliary prime field, we form the plus modular-symbol space and the winding element. The auxiliary prime q is greater than p and is 1 modulo p-1; both conditions are checked. In particular q does not divide 6(p-1).

The character and plus projectors are therefore defined integrally at q on torsion-free relative homology. The finite-field computation is reduction of this integral model, so the rank-lifting argument below applies without a torsion or change-of-lattice assumption. The operator (T_2-2chi(2)-1)(T_2-chi(2)-2) removes the Eisenstein parts. If the resulting winding vector generates a Hecke module of the full cuspidal dimension, chi is non-strange. A scalar Krylov sequence of length twice that dimension with full linear complexity proves this rank statement. Reduction of integral modular-symbol matrices cannot increase rank, so this proves non-strangeness in characteristic zero. Galois conjugacy covers every character of the same order. Random choices only search for a certificate.

The implementation either computes the sequence in Magma, or exports the same exact sparse Hecke matrix and evaluates the sequence in C++. The latter implementation checks 64 terms against Magma for each export. An entire sequence at p=176797 was also checked against its native Magma counterpart. The optional compact exporter stores the same Manin quotient map in dense/sparse blocks and constructs Hecke rows directly from Magma’s Heilbronn action. It changes neither relations nor basis. Its entire Hecke matrix was compared with the standard Magma matrix for characters of orders 2 and 29466. Every arithmetic accumulator has an explicit overflow bound. Magma computes Berlekamp--Massey; Sage independently constructs a coprime Pade certificate by polynomial Euclid and verifies its congruence, degrees and gcd. This proves minimal complexity without trusting Berlekamp--Massey alone.

All these tests pass. Consequently, v_r(str(p))<a-b, hence v_r(m/str(p))>b. Thus m/str(p) cannot divide 12. Corollary 7.5 excludes these 92 primes.

3. RATIONAL-CUSP REDUCTIONS AT 2

The remaining levels are
41,47,53,59,61,67,71,73,79,83,89,97,101,103,107,109.
The level 43 is already an occurrence.

We use the endomorphism version of the DKSS formal immersion criterion, on intermediate Jacobians. The maps are actual integral endomorphisms with finite rational image, composed with t_q=T_q-<q>-q for odd q, in DKSS’s mu-level convention. This factor kills rational torsion, hence the full maps annihilate the rational Jacobian group. For geometric point-level correspondences in Section 4, the Fricke-conjugate factor is instead T_q-1-q<q>; DKSS explicitly explains this change of modular interpretation. This avoids an assertion about an optimal winding quotient’s cotangent lattice at 2. For 107 and 109 one can take (<r>-1)t_3 directly: both primes are non-strange, so (<r>-1)J_1(p)(Q) is finite.

The selected diamond indices, and normalized multiplicity tuples covered, are
41:20,11543033; 47:23,40569972; 53:13,296084;
59:29,349638322; 61:15,961110; 67:11,79461;
71:5,419; 73:12,156334; 79:13,296084;
83:41,10495087090; 89:11,79461; 97:4,127;
101:5,419; 103:17,2784103;

107:53,152538845733; 109:54,186230004617.

At 41,61,67 and 73 several endomorphism maps are combined. Every tuple has total 12 and is normalized by a diamond translate so that its first entry is maximal; every other entry is at most 6. The corresponding Hecke/diamond operator prefixes are independent mod 2. Where the binary relation space is small, all its nonzero elements are enumerated instead: a relation with required first prefix m0 and other required prefixes m_i can occur in a normalized tuple of degree 12 if and only if max(m0,m_i)+sum_i m_i<=12. Every relation has larger cost. Otherwise all normalized tuples are tested by the C++ prefix checker. An independent dynamic program checks their number. Compressing the operator rows is exact: every original row is reconstructed from its compressed coordinates.

The tests with smaller indices 6 at 73 and 4 at 89 fail in degree 12. They are not certificates. The successful maps use indices 12 and 11, respectively. The selected files are recorded in the final ledger and formal_immersion_final.json.

4. ALL ORDINARY REDUCTIONS

The complete census over F_(2^d), 1<=d<=12, enumerates every j-invariant up to Frobenius, every twist, and every rational p-torsion line. The group order is checked not to be divisible by p^2. Sutherland coordinates identify geometric points modulo their elliptic-curve automorphisms; a Frobenius-invariant key identifies closed points. Diamond translates cover every point in an orbit. All ordinary cases consist of one closed point P and rational cusps. The census also records the supersingular cases treated below.

Let sigma denote the inverse geometric point diamond associated to the recorded primitive root. Use the integral endomorphisms
alpha=1 for 41,47,59,71;
alpha=sigma^6-1 for 61;
alpha=(sigma-1)(sigma^4-sigma^2+1) for 97;
alpha=(sigma-1)(sigma^5+1) for 101;
alpha=sigma-1 otherwise.
Their rational images are finite: the rank-zero cases and the strange-character Table 2 justify these choices. The operator A_3 kills rational torsion and actual rational-cusp divisors. Thus any ordinary closed point occurring in the reduction would have alpha*A_3(P) principal, irrespective of its rational-cusp complement.

The canonical-space construction uses all newforms, not only the rank-zero quotient. Saturation of their truncated integral q-series gives the full Z_2 lattice: if a missing integral differential existed, a minimal power of 2 and the vanishing bound for canonical sections would contradict the independence of the reduced basis. Here precision minus 1 exceeds 2g-2. Fricke transport identifies the Tate point on E(q^p) with u=q with the cusp-form expansion on E(q). The Sutherland-unit relations are global, because their verified zero orders exceed 2g-2 plus the entire pole degree. The old unit pole bounds were independently audited; the new 107 and 109 bounds are checked at every Tate cusp and are (150,157), (156,163). The geometric diamond convention is certified by the unit relations at a translated Tate point. It is not inferred from matrix order.

Ordinary local jets use a6 as parameter in

y^2+xy=x^3+a2*x^2+a6. Since j=1/a6 this is a uniformizer. The curve and p-division derivatives used in Hensel lifting are checked to be units. Actual 3-isogenies, residue degrees and trace multiplicities give the Hecke targets. Full canonical coordinates identify shared support and permit all cancellations before applying conditions.

For completeness, the nonprincipality certificate is the following elementary necessary condition. Write alpha*A_3(P)=U-V with disjoint effective U,V. Let R=H^0(K^s), with s=2 at 41,47,53 and s=1 otherwise. Its dimension exceeds deg U. Choose nonzero F in R(-U). If U-V=div(g), then S=F/g is a nonzero section of R(-V), and
 H*S belongs to F*R for every H in R(-U).
These are homogeneous linear conditions on S. A full-rank condition matrix rules out such S, and therefore rules out principality. Multiplication tensors are reconstructed from q-products beyond the relevant vanishing/Sturm bounds and every product is verified. Evaluations and multiplicities use actual jets of the closed points. The displayed F and chosen H sections are retained with each matrix.

There are 298 ordinary source orbits, all excluded. The counts at the levels in increasing order are
45,27,26,27,21,24,20,11,10,25,13,9,12,14,7,7.
Every final obstruction matrix was also checked for full rank using Sage’s independent finite-field linear algebra.

5. SUPERSINGULAR AND NONRATIONAL-CUSP REDUCTIONS

The only relevant supersingular degrees are
 41: two degree-5 points;
 73: four degree-6 points in one diamond orbit;
 109: degree-9 and degree-12 diamond orbits of size 3.
The only nonrational cusp degrees at most 12 are 10 at 41, 9 at 73, and 11 at 89. These are the orders of 2 in the diamond group. The census and these degrees also show there are no mixed supports omitted from the following list.

At 41, the independently audited native function-field certificates show A_3(P) nonprincipal for a degree-5 point and A_3(P+Q) nonprincipal for both unordered pairs up to diamond symmetry, including P=Q. They apply with rational-cusp complements of degrees 7 and 2. The degree-10 nonrational-cusp divisor is also excluded: A_3 of either such cusp is twice their difference, proved nonprincipal. Its rational-cusp complement has degree 2.

We spell out the stabilizer lifting argument, including its degree issue. Suppose h has order m, (h-1)J(Q) is finite, and the nonrational part P of the reduction of D is fixed by h. Lift the rational-cusp complement to a rational cusp divisor C and set Z=D-C. Its degree need not be zero. Nevertheless y=[hZ-Z] is a degree-zero class, and
 m*y=(h-1)[mZ-(1+h+...+h^(m-1))Z].
The class in brackets has degree zero, so the right side is torsion. Thus y is torsion. It reduces to zero. Good reduction at 2 implies 2y=0. If m is odd and h acts trivially on J(Q)[2], the identity (1+h+...+h^(m-1))y=0 also gives my=0, hence y=0.

If y=0, and gon_Q(X)>12+deg C, the principal divisor hD+C-D-hC must be zero. If only 2y=0 is known, it suffices that gon_Q(X)>2*(12+deg C). The cuspidal and noncuspidal supports are disjoint, so hD=D and hC=C separately. The rational cusps form a free

diamond orbit; consequently m divides deg C whenever C is nonempty.

At 73 use h=<2>, of order 9. The independently audited integral homology computation on the actual J_1(73), rather than an isogenous factor, proves that h acts trivially on J_1(73)(Q)[2]. The gonality bound is greater than 26. A supersingular degree-6 point plus six rational cusps would require 9|6; a nonrational degree-9 cusp plus three rational cusps would require 9|3. Both fail. If the reduction is a sum of two degree-6 supersingular points, h fixes D itself.

At 109 use h=<23>, of order 18 in the diamond group. The fresh stabilizer calculation proves it fixes the degree-9 and degree-12 closed source points. Commutativity with every diamond translate extends this check to their entire orbits, as it does at 73. Non-strangeness makes (h-1)J(Q) finite. The gonality lower bound is greater than 58. For the degree-9 point plus three rational cusps, the doubled-divisor bound is 30 and would require 18|3. For a degree-12 supersingular reduction it is 24 and forces hD=D without any rational 2-torsion assumption.

In the remaining hD=D cases at 73 and 109, a geometric diamond orbit has length respectively 9,9,3 and 18,9,6 (generic, j=1728, j=0). All orbits in an irreducible rational D have the same length. Only the j=0 shortened orbit can divide 12. Its p-subgroup is stable under the CM automorphism, so the underlying X_0(p) point has the CM field Q(sqrt(-3)) as its field of definition. As in the proof of Derickx--Stoll Corollary 7.3, the degree of its point on X_1(p) is (p-1)/3, respectively 24 and 36. Neither is 12.

At 89 a degree-11 nonrational cusp plus one rational cusp is fixed in its nonrational part by h=<2>, of order 11. The rational image of h-1 is finite. The gonality bound exceeds 39, so even the doubled divisor, of positive degree at most 26, must vanish. This would require 11|1, which is impossible.

These cases exhaust every reduction at 2, and complete the local exclusions once combined with the rational-cusp and ordinary tests.

6. STATUS AND REPRODUCIBILITY

All 603 new character tests and their Sage Pade checks passed. The 53 reused audited character certificates were rechecked as well. The final disjoint coverage verifier accepts every prime up to 532900. Its result, the complete per-prime ledger, the local matrices, and all generation and checking scripts are supplied. Failed smaller-quotient experiments are not proof inputs.

The portable packet includes the relevant mdmagma library, the borrowed exact certificates, and instructions for checking stored certificates or regenerating them in Magma V2.30-1 and Sage.

Primary references:

Derickx--Stoll, Prime order torsion on elliptic curves over number fields, Part I: Asymptotics, arXiv:2505.14109, especially Proposition 4.1, Remark 4.4, Table 2 and Corollaries 7.3 and 7.5.
Derickx--Kamienny--Stein--Stoll, Torsion points on elliptic curves over number fields of small degree, arXiv:1707.00364.
Derickx--van Hoeij, Gonality of the modular curve X_1(N), J. Algebra 417 (2014), 52--71, arXiv:1307.5719.

Parent, Bornes effectives pour la torsion des courbes elliptiques
sur les corps de nombres, J. reine angew. Math. 506 (1999), 85--116.